

Distance Control Light

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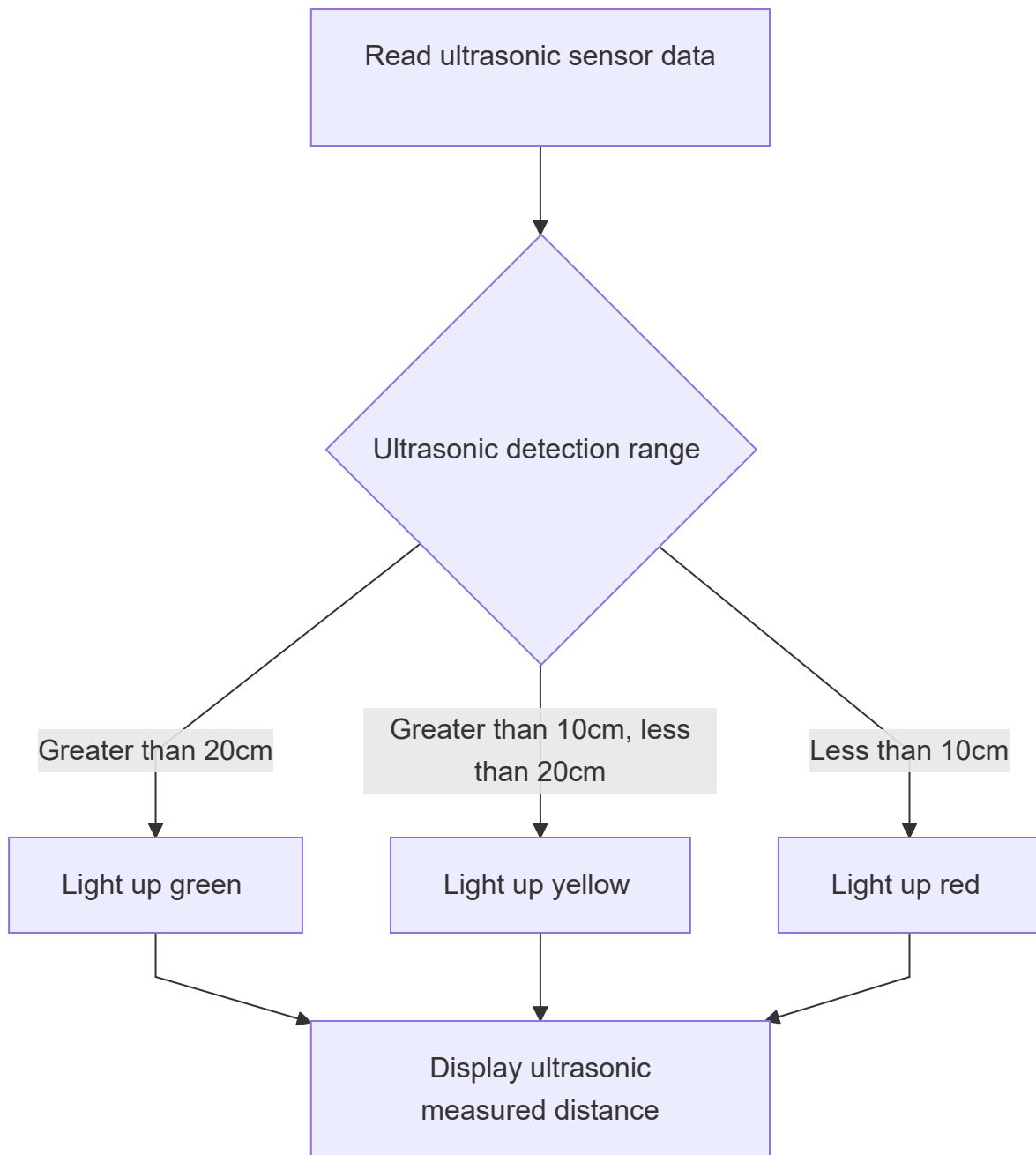
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Function Description

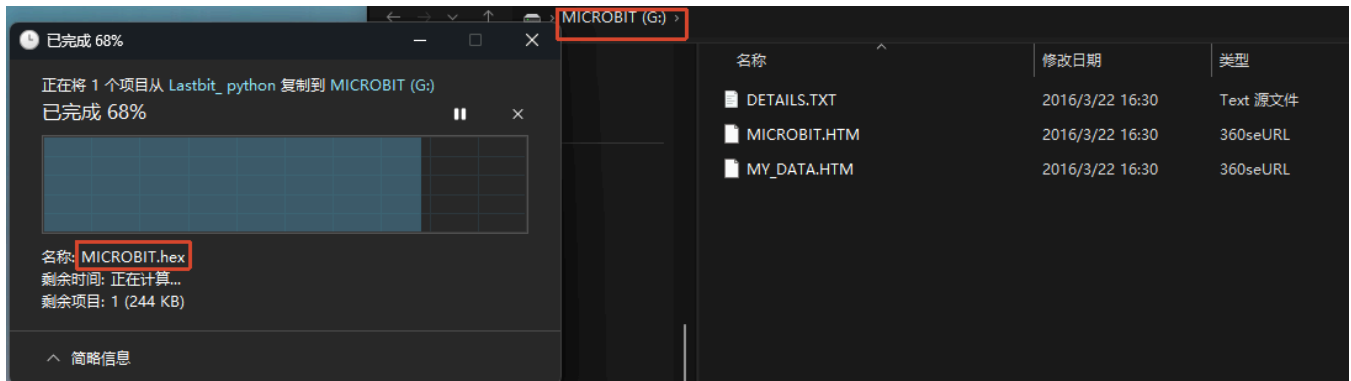
In this section, we use the ultrasonic sensor together with light control to implement "Distance Control Light".

Working Principle



Experiment Steps

Make sure `LASTBIT.hex` is flashed before flashing, otherwise an error will occur and the module cannot be imported



1. Before flashing, switch the switch to the OFF position and use a microUSB data cable to connect the computer and the micro:bit main board. **Note that it is the Microbit interface, not the Charging port on the expansion board.**
2. Open the Mu software. Here you can directly copy the program source code we provided; the operation steps are as follows. You can also enter the code manually in the editing window. Note! All English letters and symbols should be entered in English input mode. Use the Tab key for indentation, and the last line should end with a blank program.

Click the **New** button in the toolbar at the top left, and a file titled "untitled" will appear.



3. Copy the content of the **Code Implementation** section provided below into the current file. You will notice a red dot after the title bar, indicating that the code content has been modified but is in an unsaved state.

Whether to save or not does not affect code checking or flashing.

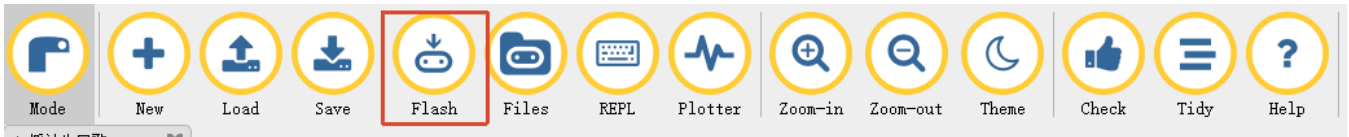
4. Now you can click **check** to verify whether there are any problems with the code format.



5. Now if you have already connected the MICROBIT to the PC in advance following the steps above, and flashed **LASTBIT.hex**, the next step is to download the current program to the Microbit.

Click the **Flash** tool in the toolbar to start flashing!

During flashing, the orange-yellow indicator light near the Microbit main board will blink frequently. When the editor shows **Copied code onto micro:bit.** at the bottom, it means the flashing is complete, and the indicator light will no longer blink after flashing.



6. Unplug the microUSB data cable, place the car on the line-tracking track, and switch the switch to the ON position.

Code Implementation

```
# 18. Touchless Light
from microbit import *
from lastbit import LASTBIT

Lastbit = LASTBIT()

display.show(Image.HEART)

# Initialize servos
Lastbit.set_all_servos(90)
sleep(200)
Lastbit.set_servo('S2', 150)

while True:
    # Read ultrasonic distance
    distance_mm = Lastbit.ultrasonic_distance()

    if distance_mm >= 200:
        # Far distance, green light
        Lastbit.all_lights_set_color(0, 255, 0)
    elif distance_mm < 200 and distance_mm >= 100:
        # Medium distance, yellow light
        Lastbit.all_lights_set_color(100, 100, 0)
    elif distance_mm < 100:
        # Close distance, red light
        Lastbit.all_lights_set_color(255, 0, 0)

    # Display distance
    if distance_mm > 0:
        display.show(distance_mm // 10) # Show in cm

    sleep(100)
```

Experimental Phenomenon

After flashing the program, the ultrasonic sensor will detect the distance to the object ahead, and different colored lights will be lit according to the different distances of the object.

When the object is far away directly in front of the car, the green light is on.



When the object is close directly in front of the car, the red light is on.

